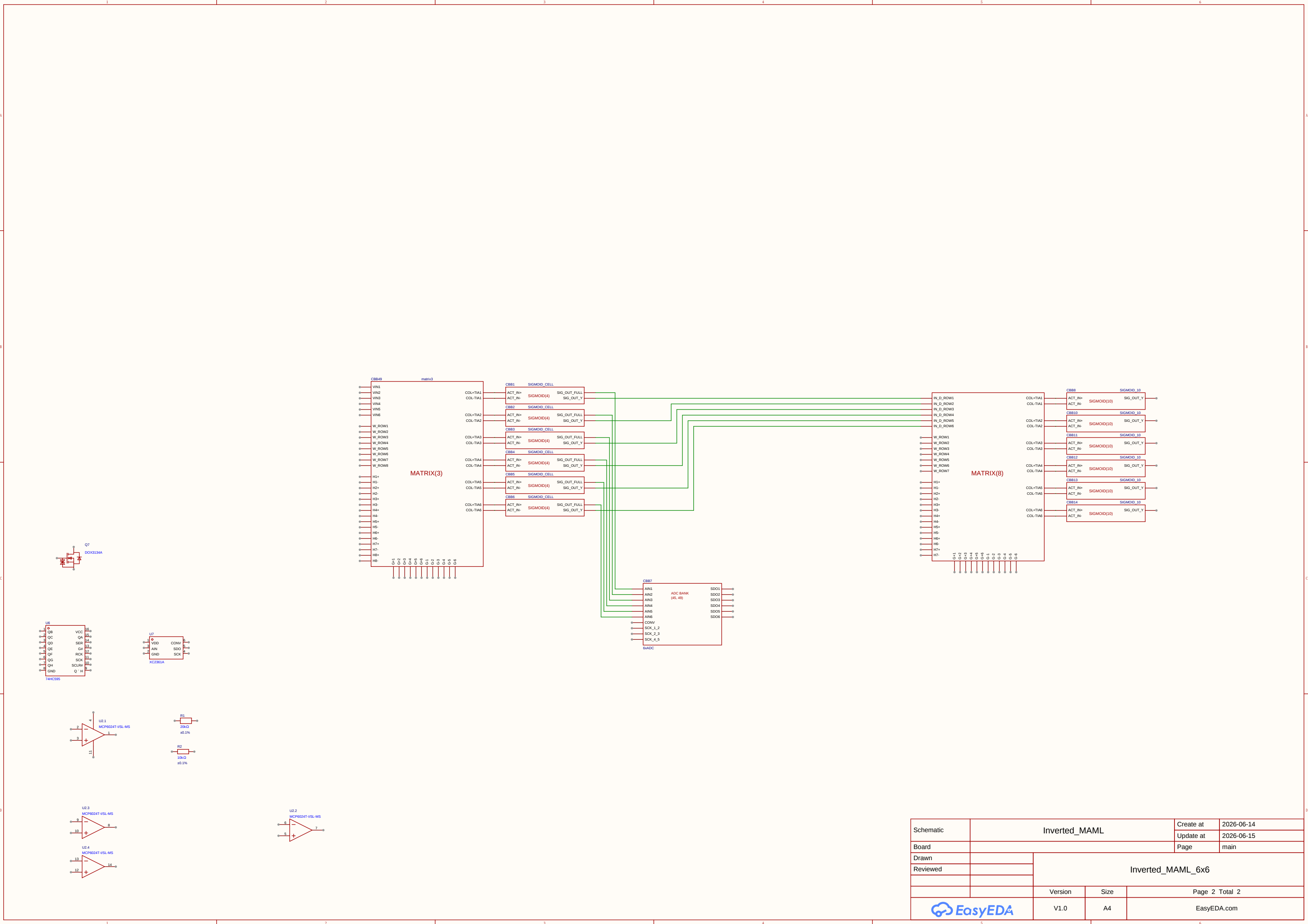
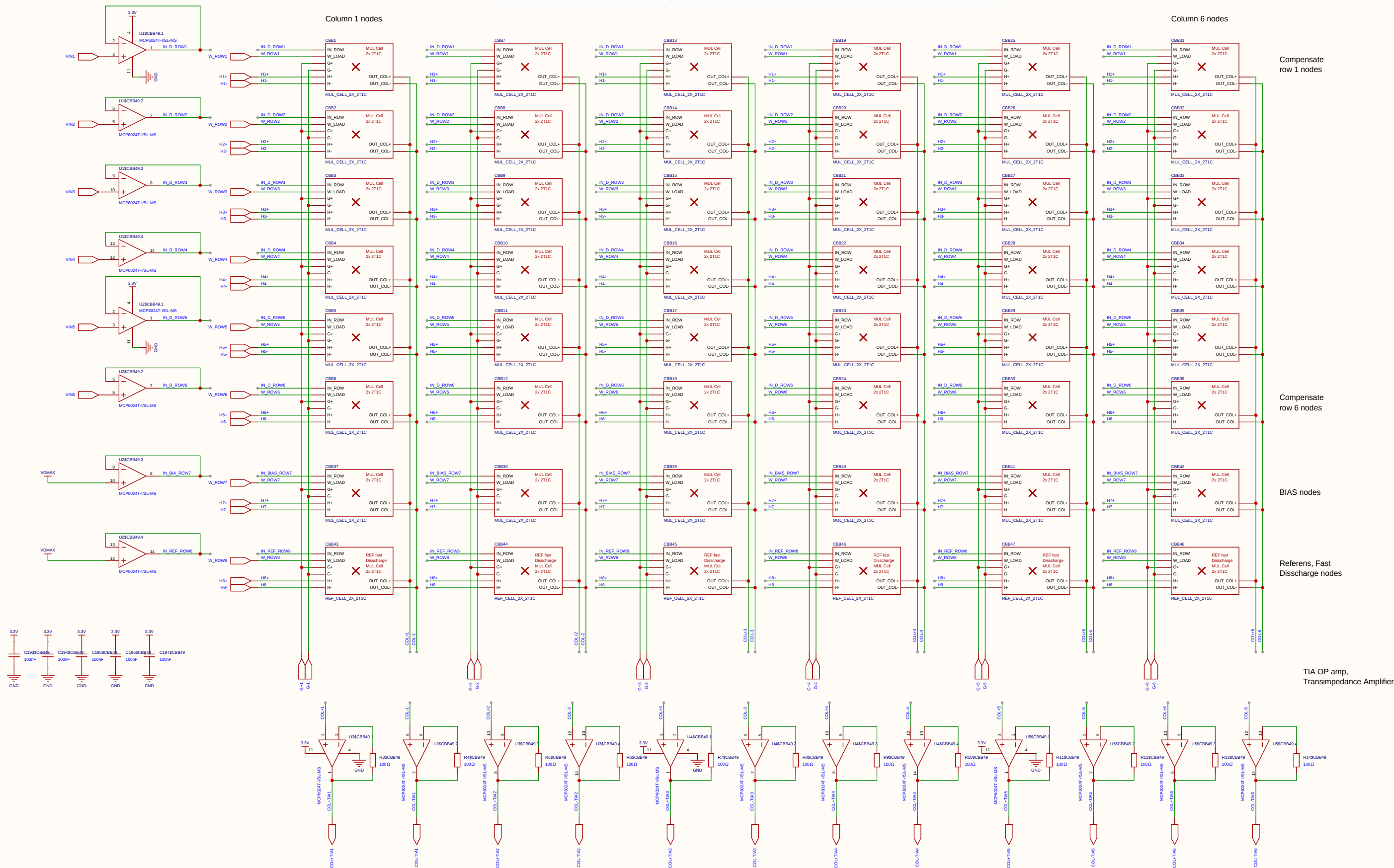




Schematic	Inverted_MAML			Create at	2026-06-14
				Update at	2026-06-15
Board				Page	main
Drawn		Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 2 Total 2	
		V1.0	A4	EasyEDA.com	

$$V_{ut_TIA} = 1.65\text{ V} - (5\text{ mA} \times 100\ \Omega) = 1.65\text{ V} - 0.5\text{ V} = 1.15\text{ V}$$



Schematic	matrix3			Create at	2026-06-14
				Update at	2026-06-15
Board				Page	matx3
Drawn		Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
		V1.0	A4	EasyEDA.com	

Category	Parameter	Value / Type	Technical Description & Function
Resistors	Discharge resistor (R)	1 M Ω	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

Component	Value	Description / Function
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a $\sim 1/101$ voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μ s.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-14
Board				Page	matrix_3_8_MUL_CELL
Drawn		Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
		V1.0	A4	EasyEDA.com	

The schematic diagram illustrates a 2D array driver circuit. It features two main sections: a column driver on the left and a row driver on the right. The column driver section includes a PMOS transistor (Q6) and an NMOS transistor (Q5) forming a current source. The row driver section includes a PMOS transistor (Q9) and an NMOS transistor (Q8) forming a current source. The circuit is driven by signals IN_ROW, W_LOAD, H+, and H-. It includes various capacitors (C1, C2, C3, C4) and resistors (R1, R2) for timing and signal conditioning. The output signals are OUT_COL+ and OUT_COL-.

Category	Parameter	Value / Type	Technical Description & Function
Resistors	Discharge resistor (R)	1 M Ω	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

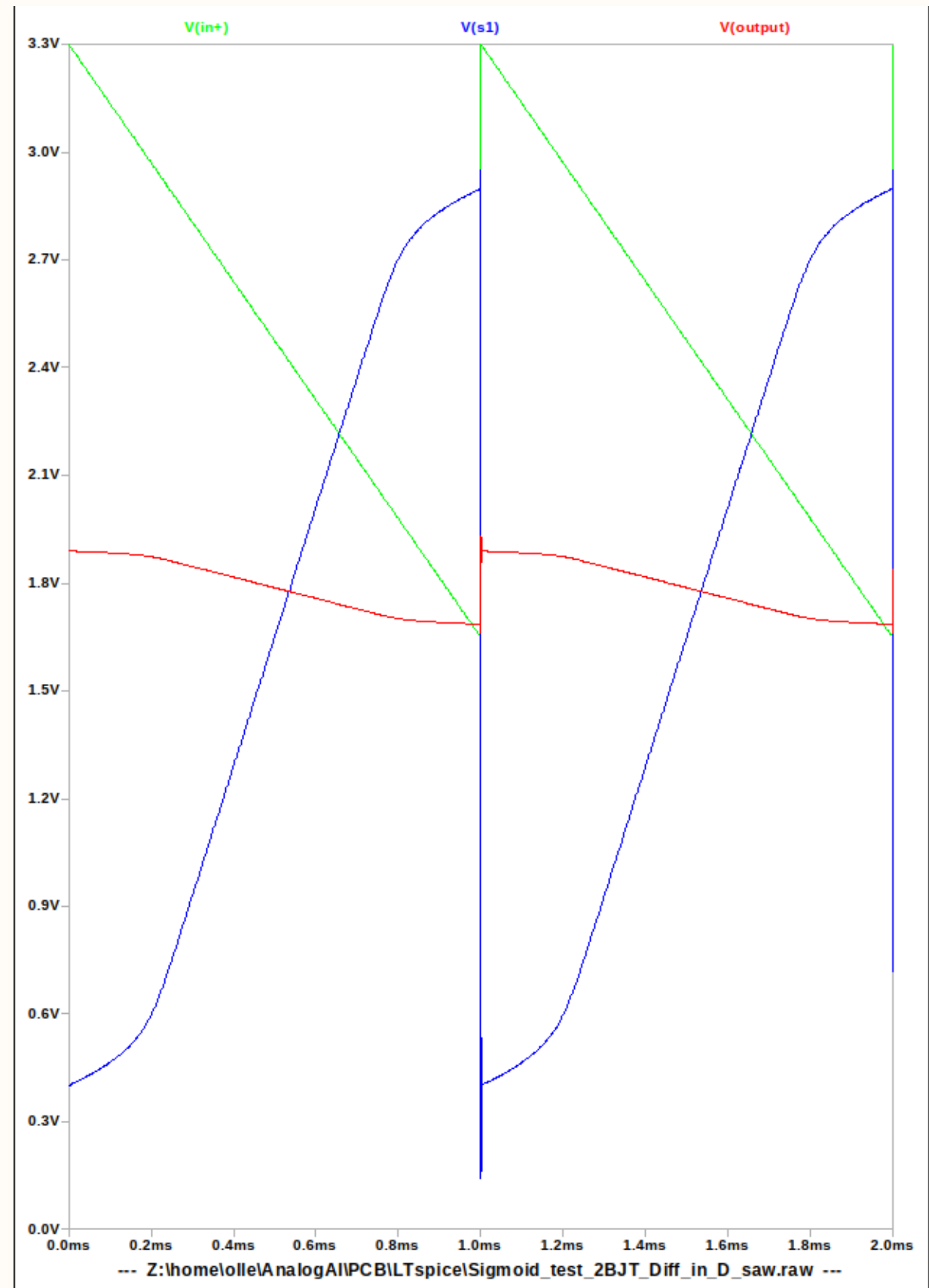
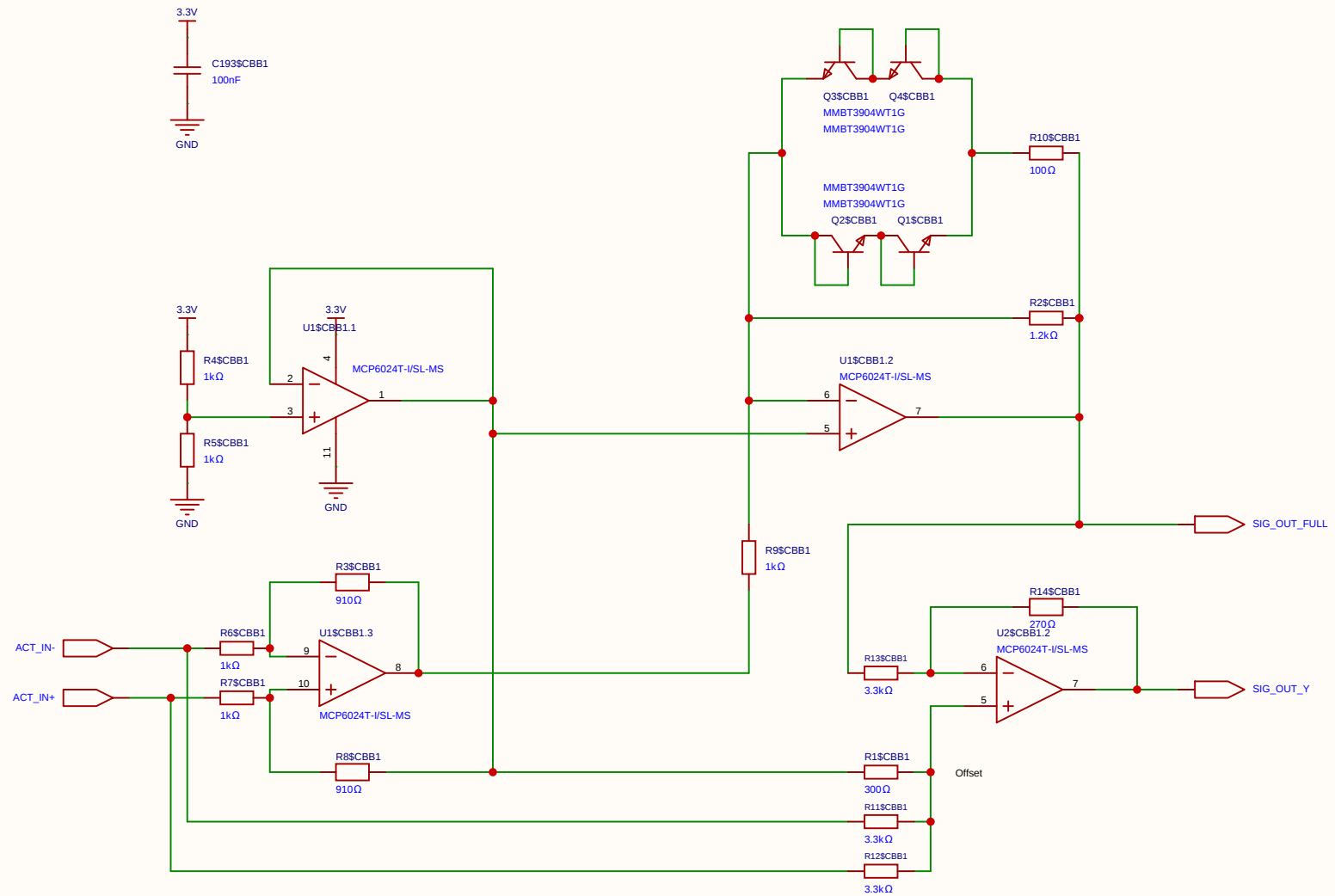
Component	Value	Description / Function
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a $\sim 1/101$ voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

State	PL Voltage	Perturbation at Gate	Comment
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μ s.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

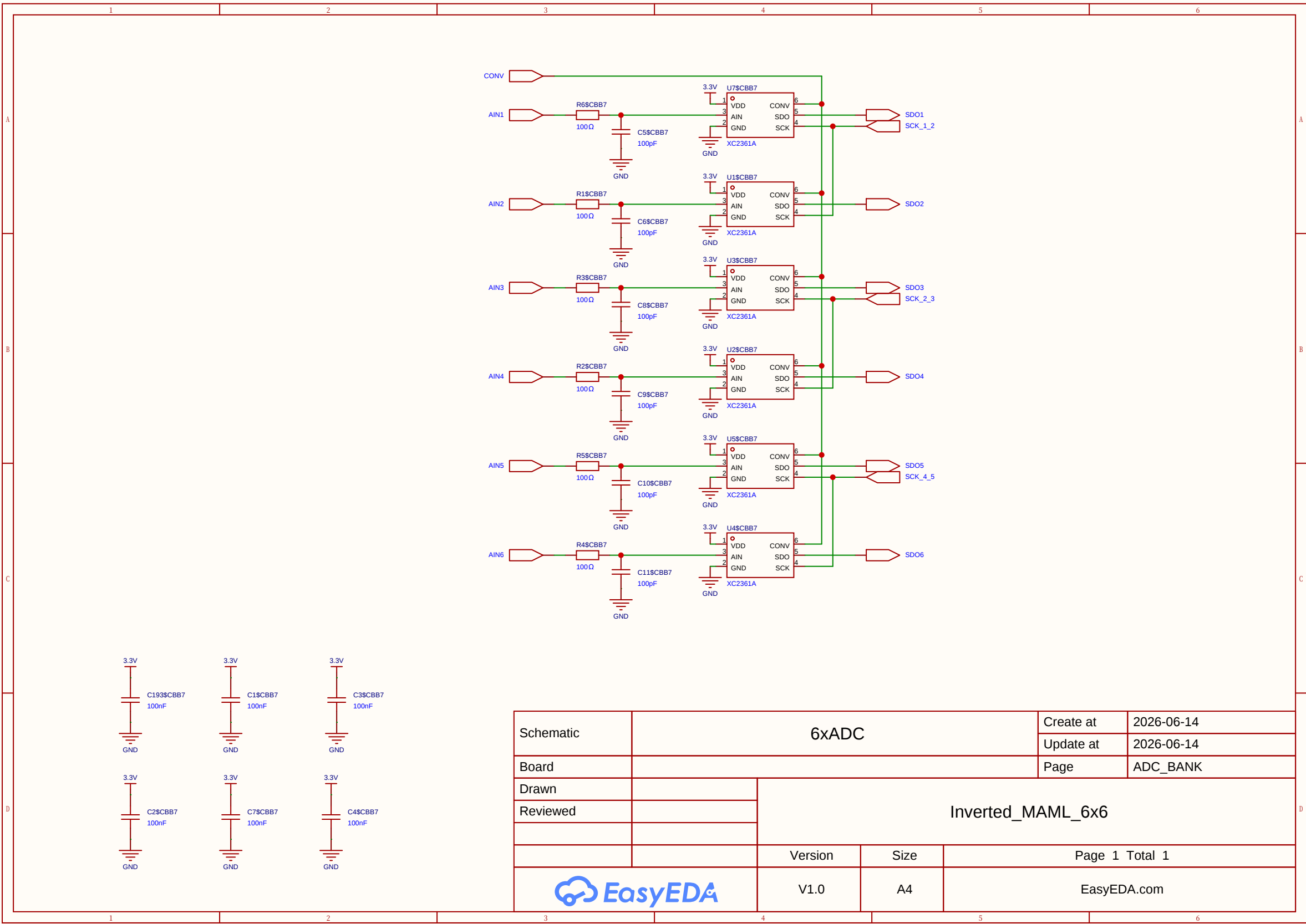
This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

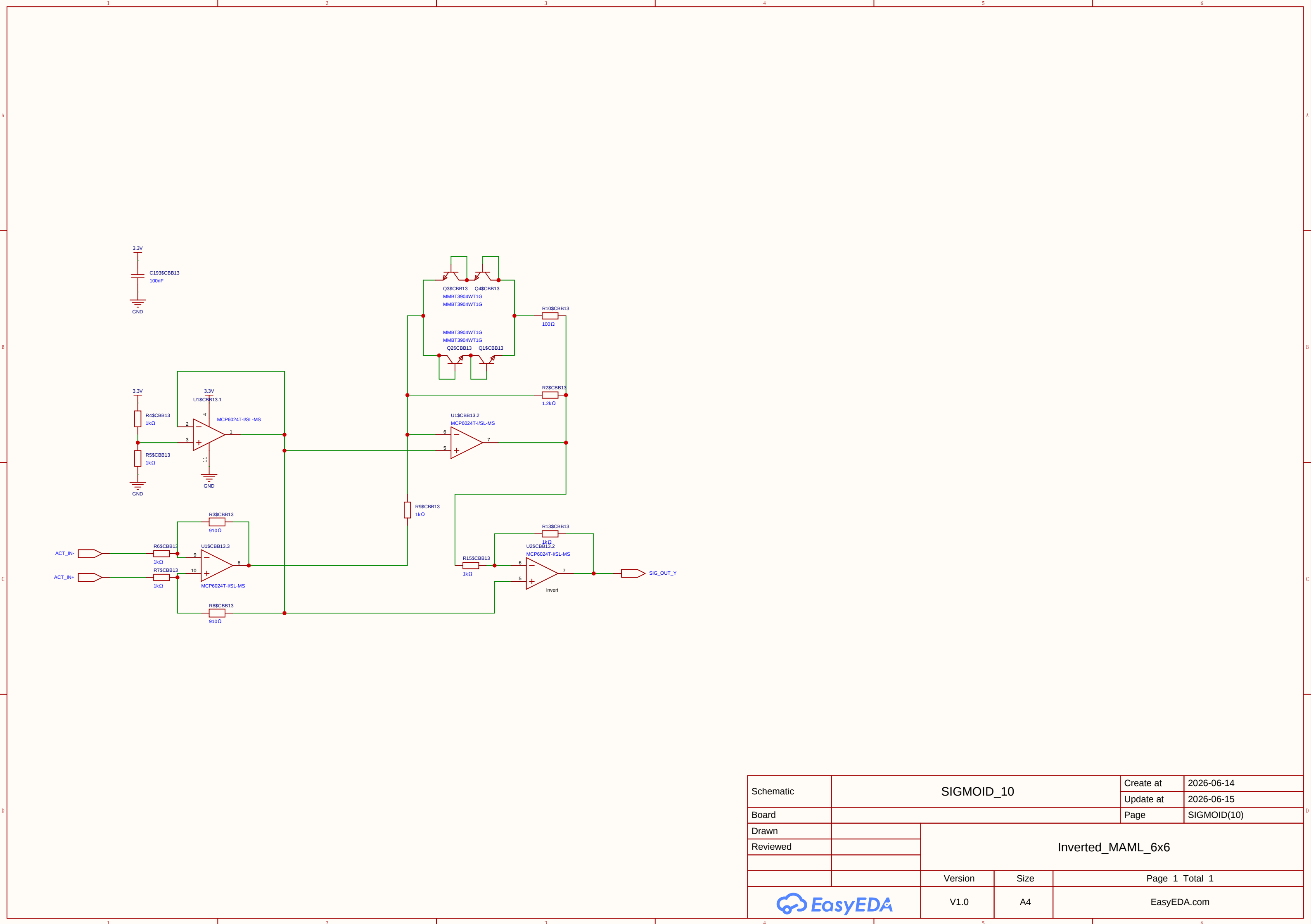
Component	Value	Description / Function
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	REF_CELL_2X_2T1C			Create at	2026-06-11
				Update at	2026-06-11
Board				Page	matrix_3_REF_CELL
Drawn		Inverted_MAML_6x6			
Reviewed					
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		V1.0	A4	EasyEDA.com	

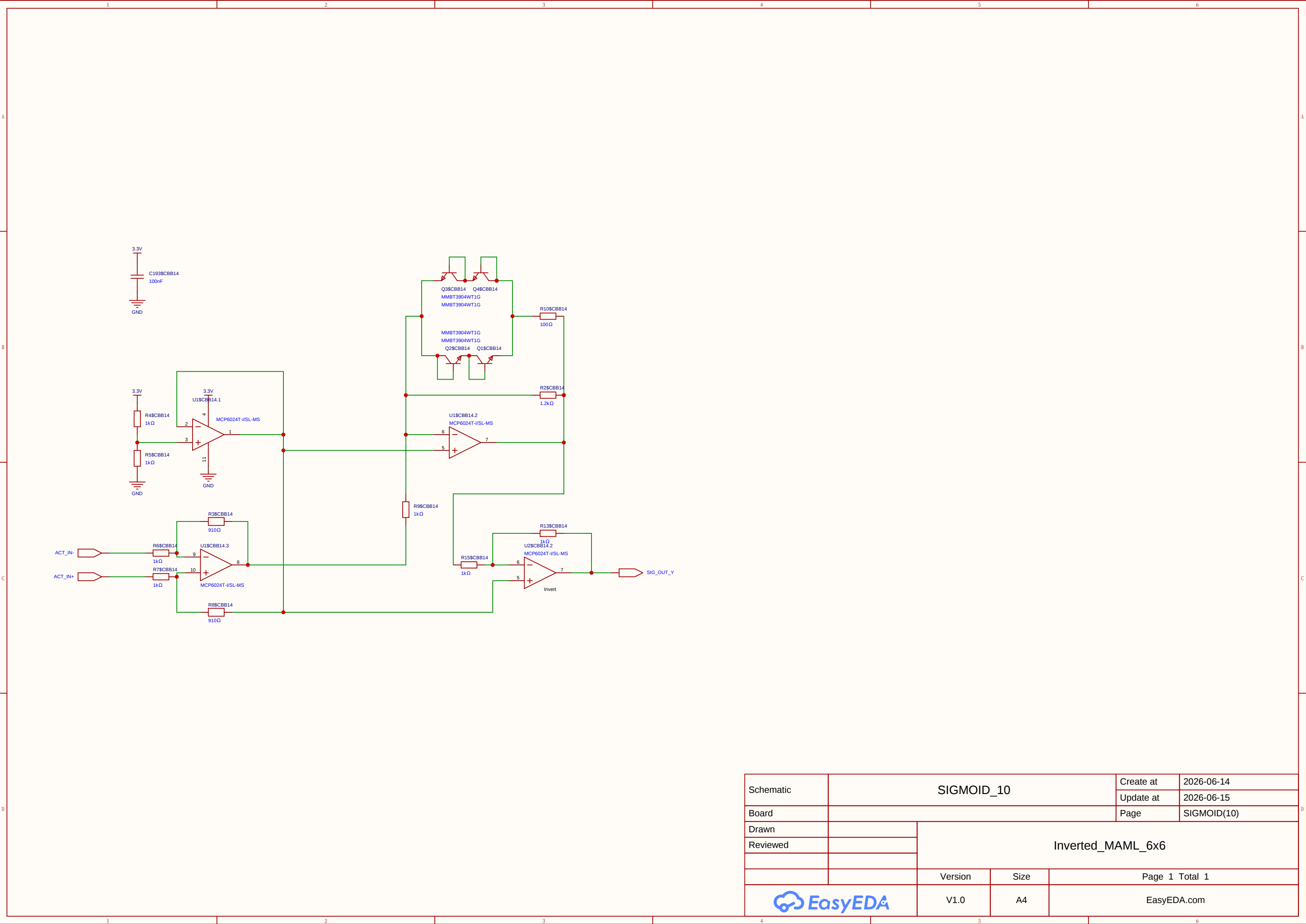


Schematic	SIGMOID_CELL			Create at	2026-06-14
				Update at	2026-06-15
Board				Page	SIMOID(4)
Drawn		Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
		V1.0	A4	EasyEDA.com	





Schematic	SIGMOID_10			Create at	2026-06-14
				Update at	2026-06-15
Board				Page	SIGMOID(10)
Drawn		Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	



Schematic	SIGMOID_10			Create at	2026-06-14
				Update at	2026-06-15
Board				Page	SIGMOID(10)
Drawn		Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

$$V_{ut_TIA} = 1.65\text{ V} - (5\text{ mA} \times 100\ \Omega) = 1.65\text{ V} - 0.5\text{ V} = 1.15\text{ V}$$

Column 1 nodes

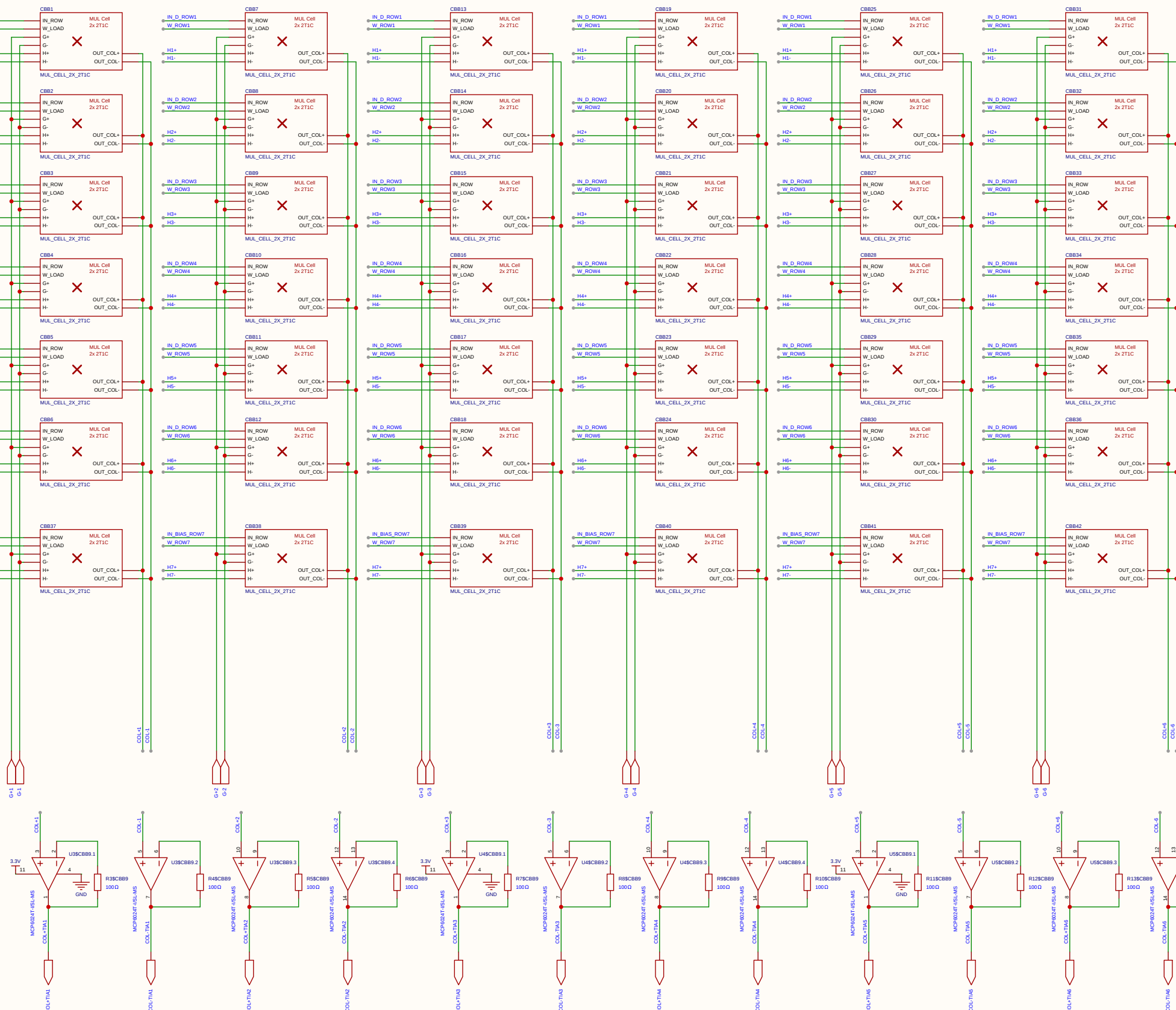
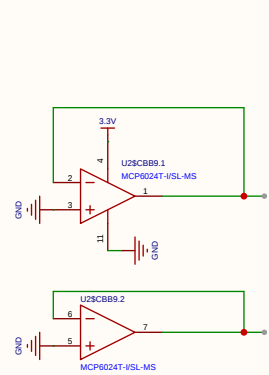
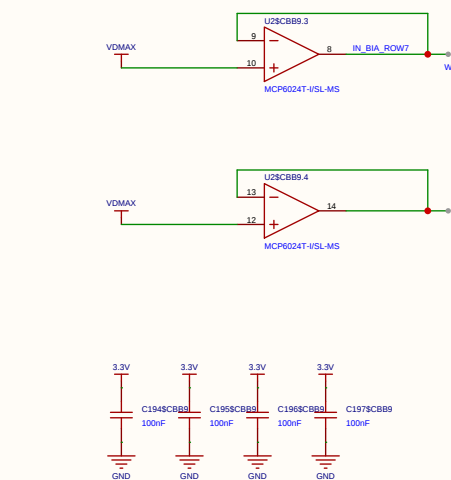
Column 6 nodes

Compensate
row 1 nodes

Compensate
row 6 nodes

BIAS nodes

TIA OP amp,
Transimpedance Amplifier



Schematic	matrix8			Create at	2026-06-15
				Update at	2026-06-15
Board				Page	matx8
Drawn		Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
		V1.0	A4	EasyEDA.com	